

GIS BASED
REJUVENATION STRATEGY
OF
RAINFED JAMANKHAL STREAM
IN THE LESSER HIMALAYAN REGION
DISTRICT DEHRADUN
(UTTARAKHAND)
(Towards Aviral Ganga)

Phase-I
Rejuvenation of Springs of
Kotimychek and Thana Danda Villages
of the Jamankhal Watershed

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Executive summary

The Himalayan region, known as the "Water Tower of Asia," is experiencing a severe crisis both in glacial-fed and non-glacial rain fed river watersheds. Rain fed rivers in Uttarakhand play a vital role in supporting agriculture, industry, and human consumption, having over 353 significant rain fed rivers critical for local communities. Within the watersheds of these rainfed rivers, about 70% villages and more than 80% urban centers of the Uttarakhand state are located. Due to anthropogenically and technogenically accelerated climate change impacts, the water sources such as springs, streams are steadily transforming from their perennial nature to ephemeral nature, and the major rivers are dwindling very fast due to decrease in base flow during summer season. Such perturbed hydrological character in the region is very critical for local communities as the environment, rural livelihoods, agriculture, and ecosystems are being adversely affecting. Failing to take a planned scientific action for groundwater augmentation through massive rainwater harvesting mechanical and biological measures in the recharge areas of springs and streams -will lead to dire consequences, including severe water scarcity, biodiversity loss, and exacerbated climate change impacts.

How to rejuvenate the dying springs/stream and steadily dwindling rivers of the rain fed watersheds ? This report is a modest attempt towards addressing this crucial problem which aims to develop a model of scientific integrated strategy for rejuvenation of springs, streams and rivers. For this purpose, a rain fed river watershed of the Kumaun Lesser Himalaya (in district Bageshwar, Uttarakhand State), viz., the Dhargad river watershed has been employed as a natural laboratory. The fundamental objectives of this work are as follows:

- i) Identify and prioritize dead and dying naulas, springs, and streams in the watershed.*
- ii) Assess the status of springs and streams and identify potential recharge areas.*
- iii) Develop a GIScience based decision support system for rejuvenation planning.*
- iv) Implement and monitor rejuvenation measures in association with local inhabitants and Gram Panchayats to improve water availability and reduce land degradation and improve the economy of the local inhabitants.*
- v) To mitigate the adverse impacts of climate change in hydrology, in particular, and the physical and human environment in general.*

A GIS-based strategy aims to identify and rejuvenate dead and dying water sources, with objectives including the assessment of springs, prioritizing recharge areas, and community engagement in implementation. The project employs GIS technologies, remote sensing, and community involvement to develop a rejuvenation plan that includes mapping water sources and planning interventions. The rejuvenation plan provides micro-plans both mechanical treatments (like infiltration holes, infiltration trenches, bio-plugs and check dams etc) and biological treatments (such as afforestation and nursery development etc) to restore the watershed. Effective implementation of the rejuvenation measures requires collaborative efforts with local communities and government agencies to ensure sustainable practices. This plan emphasizes a phased approach to address the urgent needs of the local water resources, beginning with the experimental areas of Jatha and Pass villages. Implementing these measures will enhance groundwater availability, reduce land degradation, mitigate the process of adverse impacts of climate change and improve economic conditions for local inhabitants while promoting sustainable water management practices.

1.0 THE PROBLEM

1.1 The Crucial Problem

The Himalayan region, often referred to as the "Water Tower of Asia," is home to numerous springs, streams and rivers that support millions of people downstream. However, the rain fed river regimes of the Himalaya are facing an unprecedented crisis. Springs and streams are drying up at an alarming rate, threatening the very fabric of rural livelihoods, agriculture, and ecosystems. Uttarakhand, known as the "Land of Rivers" and "Water Tower of India," is facing a severe water crisis. The state's springs, naulas, and streams, which are the lifeline of the region, are drying up at an alarming rate. The perennial water sources are becoming ephemeral/intermittent¹, affecting the livelihoods of millions of people, agriculture, and biodiversity. The spatial distribution of rain fed rivers reveals that there are as many as 353 rain fed streams/rivers of more than fourth order, having a total length of about 5796km². About more than 45% geographical area of Uttarakhand is drained by non-glacial rain fed rivers wherein about more than 70% of villages more than 80% of urban centers are located. These rain fed rivers play a crucial role in supporting agriculture, industry, and human consumption in Uttarakhand. The major rain fed river systems of Uttarakhand are Kosi, Gaula, Saryu, western Ramganga, western and eastern Nayar etc.

1.2 Importance of Springs, Naulas and Streams

- i) **Water Security:** Springs, naulas, and streams are the primary sources of water for drinking, irrigation, and other domestic purposes.
- ii) **Ecological Balance:** These water sources maintain the ecological balance, supporting aquatic life, and influencing the local climate.
- iii) **Agriculture and Livelihood:** The water from these sources is crucial for agriculture, livestock, and other economic activities, supporting the livelihoods of rural communities.
- iv) **Cultural Significance:** Springs, naulas, and streams hold significant cultural and spiritual importance in the region.
- v) **Mitigation of Climate Change Impacts:** The water system of these sources is crucial for maintaining soil moisture, groundwater storage which is prerequisite to maintain all the ecological to mitigate the adverse impact of climate change specially increasing both temperature and Co² increase.

1.3 Causes of Decline

- i). **Climate Change:** Changes in precipitation patterns, increased temperature, and altered snowmelt patterns are affecting the water cycle.
- ii) **Deforestation and Land Use Changes:** Widespread deforestation, urbanization, and infrastructure development have disrupted the natural water cycle.
- iii) **Poor Water Management:** Inadequate water management practices, lack of maintenance, and encroachment have contributed to the decline of these water sources.
- iv) **Geological Factors:** Natural geological processes, such as landslides and erosion, also impact the water sources.

1.4 The Consequences of Inaction are Dire

If no groundwater augmentation measures for rainwater harvesting through the mechanical treatment measures were taken in the rain fed watershed, the consequences of this inaction are dire which will result in severe water scarcity, ecosystem degradation and adverse impacts of climate change impacts.

- i) Water Scarcity: Decreasing water availability affects drinking water, agriculture, and other essential needs.
- ii) Food Insecurity: Reduced water availability impacts agricultural productivity, leading to food insecurity.
- iii) Migration and Social Issues: Water scarcity forces people to migrate, leading to social and economic issues.
- iv) Ecological Degradation: Drying up of water sources affects biodiversity, ecosystem services, and ecological balance.

The rejuvenation of springs, naulas, and streams in Uttarakhand is crucial for ensuring water security, ecological balance, and sustainable development. A collaborative effort from government agencies, local communities, and other stakeholders is necessary to restore these vital water sources. This project is an attempt to address this problem in an integrated and scientific way. As the springs are the sources of many streams and rivers in rain fed watersheds, therefore, the rejuvenation plan of the Jamankhal stream is divided in two phases. In the first phase, attempt is being made to develop strategy and micro-plans for rejuvenation of springs of Kotimychek and Thana Danda villages. (Fig.1).

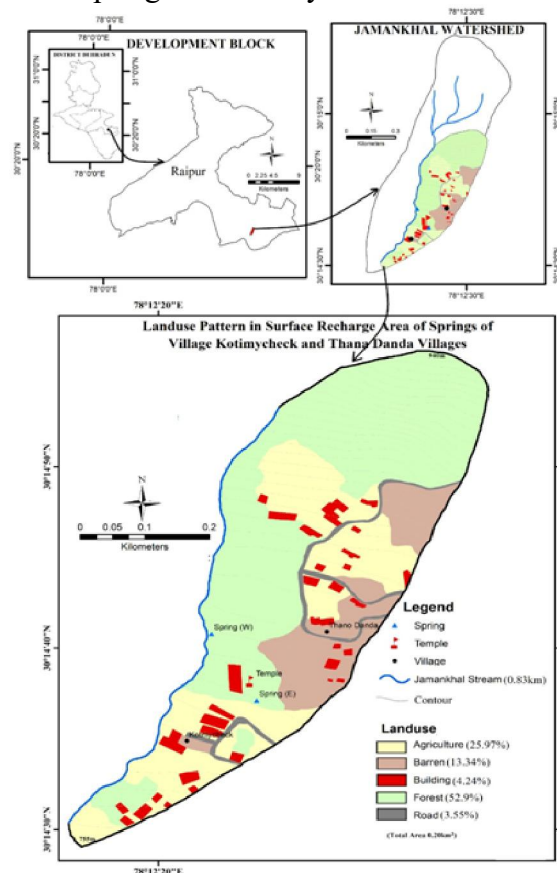


Fig.1: Location of the Jamankhal Stream Watershed.

2.0 OBJECTIVES

To address the above mentioned crucial problems, this project has been proposed to make a scientific attempt GIScience¹⁻⁴ based technologies (using GIS and Remote Sensing technologies) to rejuvenate the dying springs and streams of this rain fed experimental watershed, viz., the Jamankhal Watershed (Fig.1). The fundamental objectives of this project are as follows.

- 1. Identify and prioritize dead and dying naulas, springs, and streams in the watershed.**
- 2. Assess the current status of springs and streams, and identify recharge potential areas.**
- 3. Develop a GIS-based decision support system for rejuvenation planning.**
- 4. Implement and monitor rejuvenation measures to improve water availability and reduce land degradation and improve the economy of the local inhabitants.**

3.0 ABOUT THE JAMANKHAL STREAM WATERSHED

The Jamankhal stream watershed (Fig.1), discharges finally her water in the Song river at an elevation of 531m near Laluwala Village (30°12'44.371"N Latitude and 78°8'18.799" E Longitude,) This watershed lies in the Raipur block of district Dehradun which extends in between 30°14'28.885N to 30°15'15.457"N Latitudes and 78°12'10.52" E to 78°12'38.766" E Longitudes, and encompasses a area of about 0.54km². The altitude of this watershed varies in between 745m and 1097m

4.0 METHODOLOGY

The GIS and Remote Sensing based scientific methodology adopted in this work is divisible into four groups as presented in the following paragraphs.

4.1 GIS based Mapping of the Study Area

- i) Data Collection: Gather satellite imagery, topographical maps, and hydro-meteorological data for the Jamankhal Stream catchment area.
- ii) GIS Analysis: Delineate the watershed area, identify recharge zones, and analyze land use/land cover, slope, and drainage patterns.
- iii) Rejuvenation Plan: Develop a comprehensive plan incorporating rainwater harvesting structures, groundwater recharge measures, and watershed management practices.

4.2 GIS-based Rejuvenation Plan

- i) Recharge Zones: Identify and prioritize areas for rainwater harvesting and groundwater recharge.
- ii) Rainwater Mechanical Treatment Harvesting Structures: Propose mechanical treatment measures, i.e., construction of infiltration holes infiltration trenches, bio-plugs, check dams, contour bunds and terrace ponds etc.
- iii) Rainwater Biological Treatment Measures: Propose sites for plantation, herbal gardening and floriculture.
- iv) Watershed Management: Implement soil conservation measures, afforestation, and sustainable land use practices.
- v) Groundwater Augmentation: Design artificial recharge systems to enhance groundwater levels. Monitoring Network: Establish a monitoring system to track water levels, flow, and quality.

4.3 Implementation Strategy

- i) Community Engagement: Involve local communities in planning, implementation, and maintenance of rejuvenation measures through HESCO.
- ii) Government Collaboration: Collaborate with government agencies for technical support, funding, and policy backing through HESCO.
- iii) Phased Implementation: Implement the plan in phases, prioritizing high-impact area. This report contains Phase –I work plan.

PHASE –I **REJUVENATION OF SPRINGS OF** **KOTMYCHECKAND THANA DANDA VILLAGES** **(JAMANKHAL WATERSHRED)**

1.0 INTRODUCTION

Skilled human resource and sufficient funds are needed for reversing steady transformation springs and streams for their perennial nature to ephemeral nature. There is acute scarcity of both. In view of this acute scarcity of skilled human resource and sufficient funds, the work of rejuvenation of the Jamankhal watershed water resources has to be carried out in different phases. In the first phase –as per the need of the local inhabitants of village Kotmycheck and Thana Danda villages, a strategy has been developed by delineating recharge zone to rejuvenate the dying springs and streams of these two villages.

2.0 SITE AND SITUATION

The recharge zone of the springs of Kotmycheck and Thana Danda villages lies in between between 30°14'29.404"N to 30°14'56.521"N Latitude and 78°12'13.052" E to 78°12'34.188"E Longitude, which encompasses an area of about 0.20km² (*about 20 hectare/997 nalties*) in the Jamankha Experimental watershed (Fig.1).

3.0 PHYSICAL GEOGRAPHY

3.1 Geology

The rocks of the study are composed of Chandpur Formation of Jonsar Group (Fig.2) ⁵made-up of the following low grade metamorphic and sedimentary rocks;

- i) Quartzite – massive, well-sorted quartz grains, light-gray to white.
- ii) Sandstone – medium to coarse-grained, often cross-bedded, with occasional feldspar and mica.
- iii) Shale / Mudstone – fine-grained, thinly laminated, dark-gray to black, interbedded with quartzite and sandstone.
- iv) Limestone / Dolomite – thin to medium carbonate beds, locally recrystallized to dolomite.
- v) Slate – low-grade metamorphic rock derived from shale, with well-developed cleavage.
- vi) Phyllite – slightly higher grade than slate, with a silty sheen, found in localized shear zones.

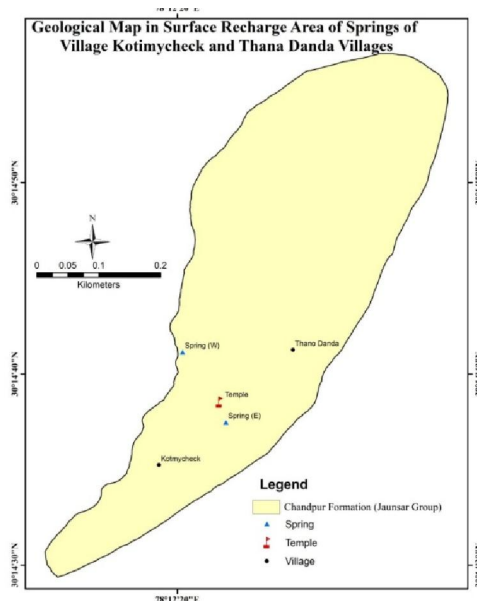


Fig.2; Geological formation of the experimental area

3.2 Pedology

There are two types of soils in the experimental area. These are alluvial soil and forest soils.

3.2.1 Alluvial Soils⁶

The middle and partially lower part of the area is capped by alluvial soils having a depth of 1.5 to 2.0m below the ground level extending down to the explored depth of 6.0-7.2m. The soil type is clay loam to silt loam and sandy clay loam in some areas. The physical and chemical characteristics of the alluvial soils are listed below

i) Physical Characteristics

Bulk Density: 1.01-1.49 Mg/m³

Particle Density: 2.14-2.52 Mg/m³

Pore Space 41.11-50.53%

Water Holding Capacity: 39.17-53.36%

ii) Chemical Characteristics

pH: Moderately acidic to slightly acidic (5.01-6.15)

Organic Carbon: 0.63-2.02%

Available Nitrogen: 177.70-696.3 kg/ha

Available Phosphorus: 16.13-29.54 kg/ha

Available Potassium: 100.64-334.92 kg/ha

3.2.2 Forest Soils⁷

The upper part of the area is covered by a thin layer of forest soils rich in organic matters where the top soil depth varies between 0/30cm and the sub-soil dept varies between 30-60 cm underlined by parent materials of weathered rocks.

i) Physical Characteristics

Bulk Density: 1.01-1.49 Mg/m³

Particle Density: 2.14-2.52 Mg/m³

Pore Space: 41.11-50.53%
Water Holding Capacity:* 39.17-53.36%
Soil Texture: Clay loam, silt loam, and loam

ii) *Chemical Characteristics*

pH: Moderately acidic (5.01-6.15)
Organic Carbon: 0.63-2.02%
Available Nitrogen: 177.70-696.3 kg/ha
Available Phosphorus: 16.13-29.54 kg/ha
Available Potassium: 100.64-334.92 kg/ha

These forest soils support lush vegetation and play a crucial role in maintaining the ecosystem

3.3 Climate and Weather

The area falls within the Sub-tropical Himalayan zone (Fig.3)⁸ which is characterized by Moderate temperatures and significant rainfall during the monsoon season. Summer temperatures range from 20°C to 35°C, while winter temperatures can drop to as low as 5°C. The area receives an average annual rainfall of 2073.3 mm, with the majority of it occurring between June and September. The average annual humidity is around 76%, with the highest levels in July and August. Sunshine. The area enjoys plenty of sunshine throughout the year.

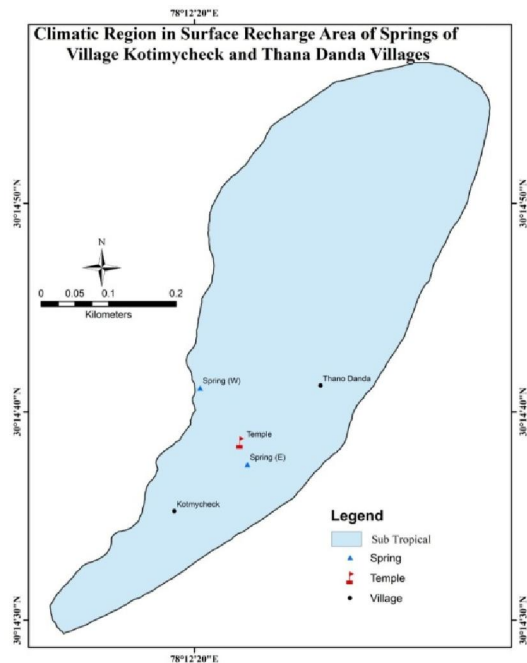


Fig.3: Sub-Tropical Climatic region.

3.4 Hydrology

Hydrologically, the experimental area is characterized by two natural springs (*locally known as Dhara*) and five ephemeral and one inermittant/perennial stream. Figure 4 Depicts the spatial distribution of these hydrological features and a brief account of the springs and streams is presented in the following paragraphs.

3.4.1 Mandir Dhara East (E)

The Mandir Dhara (E)(Plate-I) is located in the southern part of the experimental area at 30°14'37.451"N Latitude and 78°12'22.466"E Longitude, 794m elevation in village Kotmycheck. The water of this spring is used by 700 families of village Kotmuckeck and Thana Danda having a total population of 4000 persons. As informed by the local user of this spring, the water discharge capacity of this spring has gone down drastically during the last one decade. At present the water discharge capacity of this spring is found 0.71l/sec in the month of December,2025 (Table-1)

Table-1: Details of springs of th e study area.

(Source –Field Survey by HESCO,Dehradun)

SN	Water Source	Type	Beneficiaries Villages	Beneficiaries Families	Beneficiaries Population	Water Use	Condition 10 years back	Present Discharge Capacity
1	MandirDhara(E)	Spring	Kotimychek, ThanoDanda	700	4000	Drinkin g	It was good in discharge	0.7 l/sec
2	MandirDhara (w)	Spring	Kotimychek, ThanoDanda	700	4000	Drinkin g	It was good in discharge	0.04 l/sec



Plate-I: Mandir Dhara (E) located in village Kotmycheck

3.4.2 Mandir Dhara West (W)

This spring (W) (Plate-II) is located in the southern part of the experimental area at 30°14'41.115"N Latitude and 78°12'20.248" E Longitude, elevation of at and 809m in village Kotmycheck. The water of this spring is used by 700 families of village Kotmuckeck and Thana Danda having a total population of 4000 persons. As informed by the local user of this spring, the water discharge capacity of this spring has gone down drastically during the last one decade. At present the water discharge capacity of this spring is found 0.041l/sec in the month of December,2025 (Table-1)



Plate-II: Mandir Dhara (W) located in village Kotmycheck

3.4.3 Ephemeral Streams

The upper northern part of the area is carved by ephemeral streams which become very active during the rainy season. There are 5 ephemeral streams having their total length of about 0.92km

3.4.4 Jamankhal Stream

This course of the Jamalikhhal stream (*locally known as gadhera*) makes the boundary of the western part of the experimental area (Fig 11). The total length of Jamalikhhal stream is about 0.82km out of which only about 0.20km (%) in her lower reach is perennial in nature which is also in dying stage (Plate-III). The upper part of this stream is ephemeral (Plate-III) and is active during the rain only while the middle part has intermittent in nature have water flow for 4-6 months of the year (Plate-IV).



Plate-III: Jamankhal ephemeral Stream



Plate-IV: Jamankhal seasonal Stream

3.5 Geomorphology

A detailed account of geomorphology –which contains, about relief, slopes and morphological features of the experimental is presented in the following paragraphs.

3.5.1 Relief

The contour map (Fig.4) reveals that the average height of the recharge zone stands at 847m which varies in between 755m to 940m at the top of the recharge zone. The relief map (Fig.9) of the recharge zone reveals that about 21.7% (0.05km²) area of the recharge zone lies on very high relief zone where the height is more than 900m, 30.02% (0.06km²) area of the recharge zone lies on high relief zone where the height between 850 m to 900m, about 17.09% (0.03km²) lies on the moderate relief zone where the height varies in between 800m to 850m and the remaining 30.2% (0.06km²) area falls under low relief zone where the height varies between <800m (Table-2).

Table-2: Distribution of relief of the Kotimychek and Thana Danda Villages Springs and Naula Recharge zones in different relief zone.

Relief Zones in Meters	Area (in km ²)	Area in %	Remark
< 800	0.06	30.2	Low
800-850	0.03	17.9	Moderate
850-900	0.06	30.2	High
>900	0.05	21.7	Very High
Total	0.20	100	-



Fig.4: Geographical distribution of contours showing topographic structure of the study area with maximum and minimum and contour heights at 1m interval.

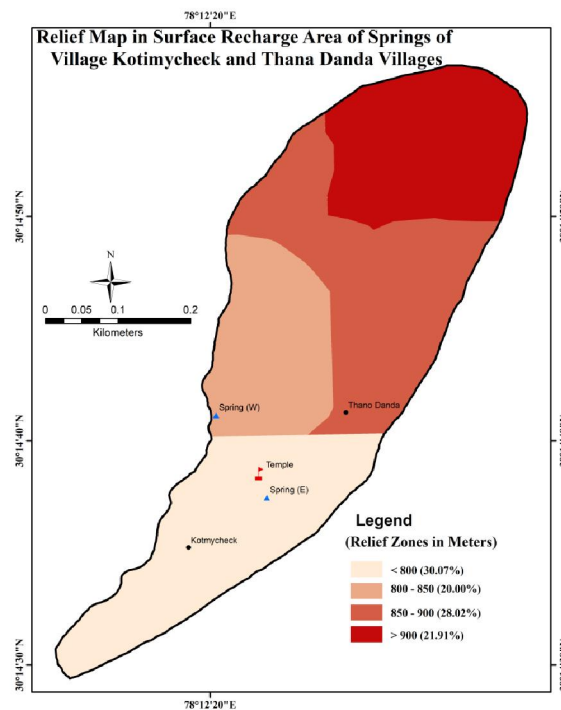


Fig.5:Relief map of the experimental area showing different relief zones.

3.5.2 Morphology

For delineating micro level geomorphic landforms, the satellite image of the study area was overlain by the high resolution contour map (Fig.7).Using this map (Fig.7), micro-geomorphic features were marked as depicted by figure 10.This morphological map, reveals that the landscape of the experimental area is mainly carved by the fluvio-tectonic (river and indogenic), technonic and pluvial (rainwater) process. A brief account of morphological features of different genetics is presented in the following paragraphs.

3.5.2.1 Landforms developed by Fluvio-tectonic Processes

The landforms developed by the fluvio-tectonic process in the region are the imprints of multi-tier river terraces, triangular fault facet and steep planner slopes in the experimental area.

i) *River Terraces*-The multi-tier river terraces (Fig.12) are the outcome fluvio-tectonic processes. There are three levels of river terraces designated as T1, T2 and T3. These terraces are bed-rock terraces which are geomorphologically known as strath⁶ and are covered by a thin layer of alluvium. These multi-tier imprints of fluvio-tectonic terraces are sharp indicators of past rapid successive uplift of this region. The area of these river terraces is about 0.92km² which accounts for about % (km²) area of the total experimental area.

ii) *Triangular Fault Facet*

The uppermost area of the experimental area is in the form of triangular facet⁷ overlooking to a fault and is caused by the differential uplift along the fault plain. This triangular fault facet is highly degraded by pluvial processes of rainwater. This triangular fault facet encompasses about % (km²) area of the total study area.

iii) *Straight Planner Slope (SP)*

At number of places there is straight planner slope designated as SP in the geomorphic map (Fig.8). Genetically, the straight planner slopes are imprints of tectonic uplift of the area.

3.5.1.2 Landforms Carved by Pluvial Processes

Besides fluvio-tectonic landform, the area is also carved by erosional and depositional pluvial process of rainwater. The outcome of erosional pluvial landforms are convex (CV) and concave (CV) slope units, and the ephemeral rills; while the imprints of depositional pluvial processes are, debris cones and debris fans. Figure 10 depicts the spatial distribution of all these landforms carved by the pluvial processes.



Fig.6: The satellite image of the study area overlain by high resolution contour (1m resolution) map

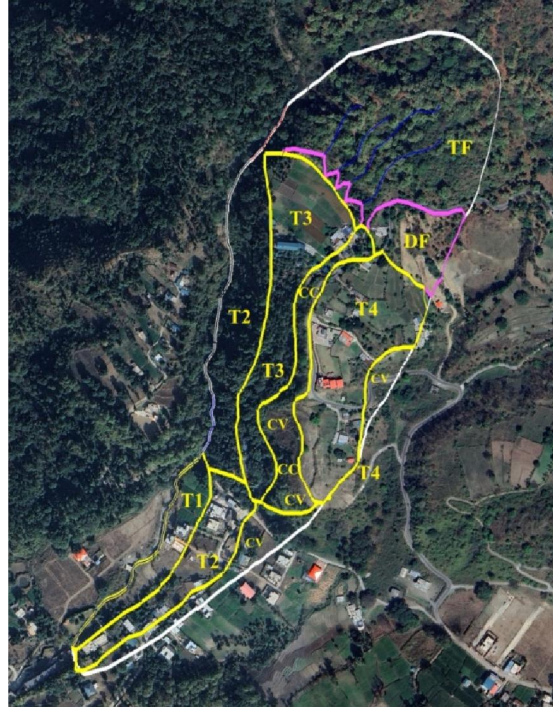


Fig.7:Geomorphic Map of the study area.

4.0 HUMAN GEOGRAPHY

4.1 Land use Pattern

The landuse pattern of the study area is presented in figure 11. Out of the total area, a large part,i.e.,about 52.9% (0.11km^2) area is under forest land.About 13.34% (0.03km^2) area falls under barren land. About 0.05km^2 area, which constitute about 7.9% of the total area is under agriculture land which is now mostly in abandoned form due to migration of people from the village. The settlements in the recharge area cover about 4.24% (0.01km^2) area while about 3.55% (0.005km^2) area is covered by road.

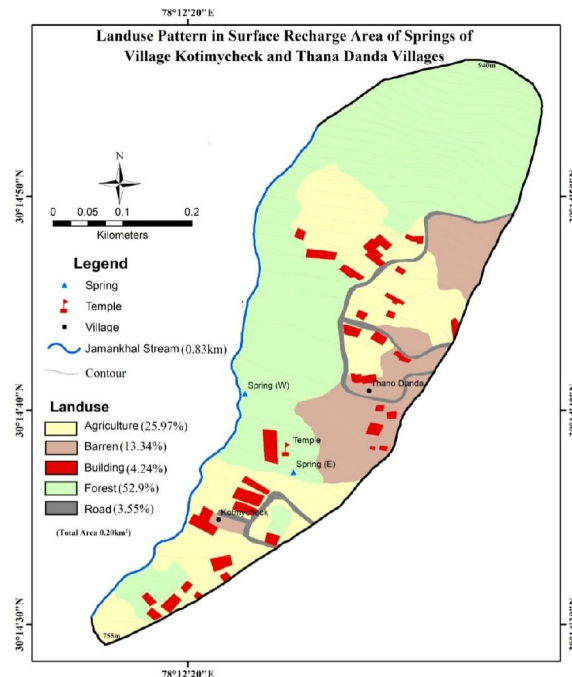


Fig.8: Land use pattern in the study area

4.2 Drinking Water Schemes

Table-4: Current drinking water (personal/jalsansthan) schemes other than above two springs
(Source –Field Survey by HESCO, Dehradun)

SN	Name of Scheme	Water Source Location	Length of Scheme	Beneficiaries Population	Water Use	Year of construction	Present Discharge Capacity
1	Swajal Gravity Scheme	Sisarukhala	5 Km	600	Drinking	Ramnagar Danda (Over 10 year)	60 ltr /minute
2	Swajal Gravity Scheme	Jamankhala	1.5 Km	100	Drinking	Kotimychek (Over 7 Year0)	15 ltr / minute

5.0 PROPOSED REJUVENATION TREATMENT MEASURES

The treatment measures to be taken for rejuvenation of the spring of the experimental area are divisible into two different types These are mechanical treatment measures and biological treatment measures. A brief account of objectives of these rejuvenation measures is presented in the following paragraphs..

5.1 MECHANICAL TREATMENT

The biological treatment measures take longer time to improve the condition of land for rain water infiltration, therefore, for speedy groundwater augmentation, mechanical treatment measures are needed in the recharge area. The objectives and types of mechanical treatments to be taken are described in the following paragraphs.

5.1.1 Objectives

Particular Objectives

- Groundwater augmentation by increasing the rainwater percolation capacity of bed rocks to recharge groundwater storage.
- To increase soil moisture storage in the top-soil, sub-soil and parent material horizons of land.
- To control surface runoff or the problem of flooding;

General Objectives

- To control accelerated soil erosion;
- To reduce dissolved, suspended, and bed load concentration of streams/rivers water.
- To bring back *haryali* on the nature through water conservation;
- To improve the quality of economic environment of the local environment.

5.1.2 Treatment Measures

To complete above objectives, the following six structures are recommended for mechanical treatment to fetch the rainwater for groundwater augmentation to recharge springs, naulas and streams. These are: infiltration holes, infiltration trenches, bio-plugs, rill/gully plug, contour bunds, terrace ponds, check dams and building roof top rainwater harvesting structures. A brief accounts of these structures is presented below.

5.1.2.1 Infiltration Holes (Plate-V)

These are small holes (size 10-15 cm deep, 5cm dia and density 5 per/m²) as shown in *Plate-I* to be constructed on the degraded forest and barren lands to catch and infiltrate the rain water of the high intensity rain storms into the soil for groundwater augmentation.



Plate-V: Example of infiltration holes (size 10-15 cm deep, 5cm dia and density 5 per/m²) constructed to fetch high intensity rain drops in barren as well as in degraded forest to facilitate infiltration of rainwater for groundwater augmentation.

5.1.2.2 Infiltration Trenches (Plate-VI)

These are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in *Plate-VI*, on the hill slopes made at a spacing of 10m at an average slope condition of 20°. The distance of spacing may be decreased, in case the hill slope is low. These should be constructed in stagger pattern. The purpose of these infiltration trenches is to catch the excess hill slope runoff and to facilitate it for groundwater recharge. These infiltration trenches should not be constructed on discharge zones and flat lands.



Plate-VI: An example of infiltration trench (size 1.0 x 0.5 x 0.3 m) to trap the overland Sheet flow from the hill slope for infiltration and also to trap the sheetwash erosion sediment.

51.2.3 .Bio- Plugs (Plate-VII)

These are small bio-plugs to be constructed by vegetation and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rain water to facilitate it for groundwater recharge (Plate-VII). The size of these bio-plugs depends upon the local terrain condition.



Plate-VII: An example of bio-Check dam constructed using pine niddle across a first order ephemeral stream to break the velocity of runoff, to store water for infiltration in to the aquifer and to ckec the erosion.

5.1.2.4 Rill Plugs/Gully Plugs (Plate-VII)

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep gullies and rills are being developed. To control this process of gullying/rilling, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of screams, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-VIII) depends upon the local terrain condition.



Plate-VIII: An example of gully plug across a first order ephemeral stream to check runoff velocity, to store the rainwater and to control process of erosion.

5.1.2.5 Contour Bunds (Plate-IX)

The contour bunds are earthen embankments along contours to capture rainwater in agricultural land (Plate-IX). These help in storing the rainwater for infiltration to groundwater augmentation, reduce erosion and also help in irrigating agricultural crops. These contour bunds should develop in association with the local residents.



Plate-IX: An example of series of contour bunds constructed on the edges of agricultural fields to trap rainwater for irrigation as well for infiltration to recharge groundwater storage.

5.1.2.6 Terrace Ponds (Plate-X)

The terrace ponds are small, artificial water tanks constructed on terraced fields in gentle hill slope areas. These are small shallow tanks with earthen embankments. Their size varies, but typically 10-20 m² in area and 1-2 m deep. The Dig pond area, using the excavated soil of the dig pond are should be used for construction of embankments. They're designed to collect and store rainwater through nearby channels or pipes bring runoff from surrounding catchment areas runoff, providing water for recharge in to the aquifer, irrigation and other uses. They also help to control erosion in the hills.



Plate-X: An example of terrace pond for rainwater harvesting to groundwater recharge and irrigation of different crops.

5.1.2.7 Check Dams (Plate-XI)

These are series of small check dams, constructed using cement and iron across second/third order streams having a height of 1.0 m to 1.5m (depending upon the local terrain condition of the site). The purpose these check dams constructed across stream at an interval of 30m to 50m (depending upon slope condition) is to check the flow of intermittent/seasonal streams to facilitate it for groundwater recharge, and to mitigate the process of erosion.



Plate-XI: An example of series of check dams across an intermittent/seasonal stream to collect water for infiltration and irrigation purposes.

5.1.3 Building Rainwater Harvesting for Domestic Use (Plate-XII)

Rainwater harvesting systems collect and store rainwater from rooftops for domestic use, reducing dependence on external water sources. This system has 4 components:

- i) Rotoff collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Distribution system: Supplies harvested water for domestic purposes (e.g., flushing toilets, washing clothes, cleaning, irrigation)

This rainwater harvesting system has the following benefits:

- i) Water conservation: Reduces water bills and dependence on external sources
- ii) Reduced stormwater runoff: Decreases burden on drainage systems
- iii) Non-potable use: Suitable for uses that don't require treated water
- iv) Eco-friendly: Promotes sustainable water management practices.

This rain water system may be used for the following domestic uses such as: i) flushing toilets, ii) washing clothes, iii) cleaning, iv) irrigation, and v) washing utensils/vehicles etc.

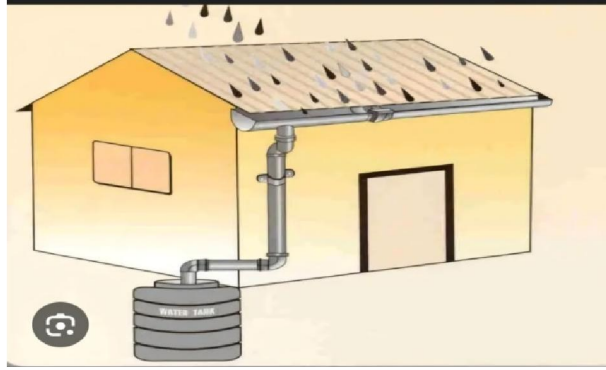


Plate-XII: Rooftop rainwater harvesting structure for kitchen gardening and utensil cleaning etc

5.1.4 Building Rainwater Harvesting for Groundwater Augmentation (Plate-XIII)

Building roof rainwater harvesting structures in hills can help rejuvenate dying springs and streams by augmenting groundwater resources. The system captures rainwater from rooftops, stores it, and allows it to percolate into the ground, recharging aquifers (Plate-IX). This system has the following 4 components (Plate-IX).

- i) Rotoff collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.

The benefits of this system are: i) rejuvenates dying springs and streams, ii) augments groundwater resources, iii) reduces soil erosion, iv) conserves water, and mitigates water scarcity in hilly areas.

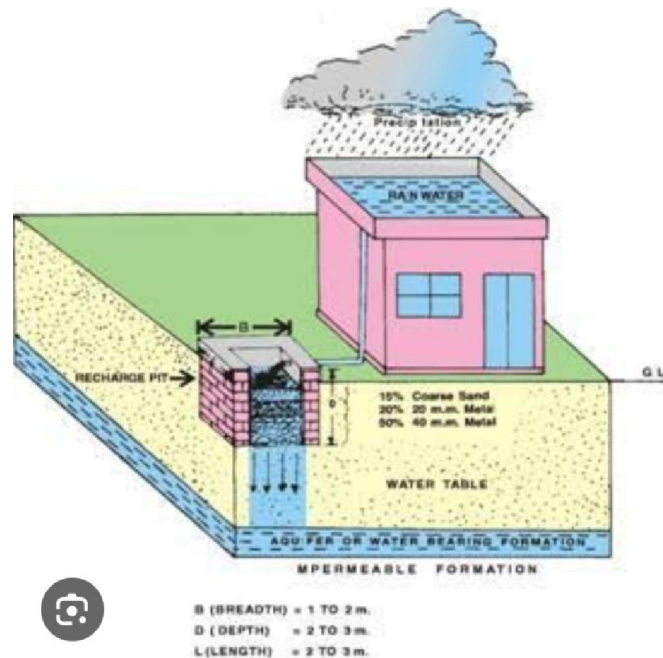


Plate-XIII: Rooftop rainwater harvesting structure for groundwater

5.1.4 Rainwater Harvesting for Groundwater Augmentation and Domestic Use (Plate-XIV)

Building rainwater harvesting structures in hills can serve multiple purposes also, including domestic use, groundwater augmentation, and rejuvenation of dying springs and rivulets. The system captures rainwater from rooftops, stores it, and utilizes it for various needs. This system has five components (Plate-XIV):

- i) Roof collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.
- v) Distribution system: Supplies harvested water for domestic purposes (e.g., irrigation, toilet flushing, cleaning).

There are different benefits of this system as mentioned below.

- i) Domestic use: Provides water for non-potable purposes, reducing dependence on external sources
- ii) Groundwater augmentation: Recharges aquifers, rejuvenating dying springs and rivulets.
- iii) Soil conservation: Reduces soil erosion and landslides
- iv) Water scarcity mitigation: Helps alleviate water scarcity in hilly areas
- v) Eco-friendly: Promotes sustainable water management practices,
- vi) Rejuvenation of Dying Springs and Rivulets: Recharged groundwater feeds into springs and rivulets, restoring their flow and
- vii) Helps maintain aquatic ecosystems and supports biodiversity

Thus, by implementing rainwater harvesting structures in hills, communities can ensure a sustainable water supply, conserve natural resources, and promote eco-friendly practices.

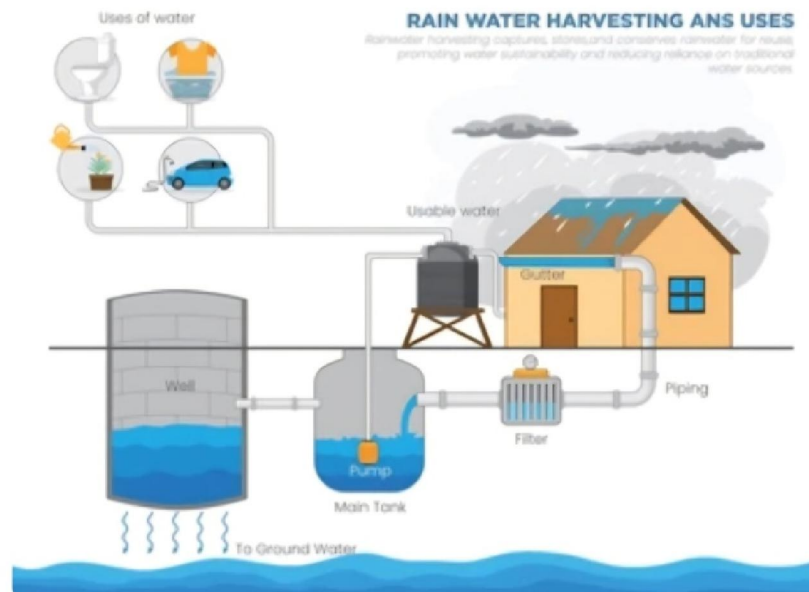


Plate-XIV: Rooftop rainwater harvesting structure for groundwater augmentation for rejuvenation of springs and for different domestic purposes

5.2 BIOLOGICAL TREATMENT

Biological treatment measures include –afforestation, plantation of trees, shrubs and grasses, and development of nurseries and horticulture. The past experiences reveal that, no biological treatment work should be done for rejuvenation of dead/dying water resources without ensuring the complete protection from fire as well as from damages through animals and human beings. A brief account of objectives and treatment measures of biological treatment is presented below.

5.2.1 Objectives

Particular Objectives

1. To increase the rainwater infiltration capacity of land for soil moisture storage.
2. To increase groundwater storage by deep percolation of rainwater into rocks on long term basis.
3. To decrease surface runoff.

General Objectives

1. To control accelerated erosion.
2. To mitigate the problem of flooding on long term basis.
3. To control the process of mass wasting, i.e., land creeping, slumping, land sliding etc,
4. To regenerate the degraded natural environment.
5. To improve the quality of economic environment of the local inhabitants..

5.2.2 Treatment Measures

In order to meet above objectives, the biological treatment measures such as afforestation, plantation, and development of horticulture and nurseries should be done. A brief accounts of all these biological treatment measures is presented in the proceeding paragraphs.

5.2.2.1 Afforestation

Afforestation means establishing a forest in an area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, promote biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in such available areas afforestation of water harvesting (broad leaved) species, local shrubs and grasses such as Napier and vertebrae should be done by initiating massive social forestry programmes in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.

5.2.2.2 Plantation

This is a process of establishing a forest plantation in an area often with a single species (monoculture). Typically, plantation involves non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department.

5.2.2.3 Horticulture

It includes the development of community horticulture and individual horticulture programmes specially on the barren and abandoned agricultural lands in the recharge areas by which the waste land of the watershed could be carpeted by plants to increase rainwater infiltration capacity of land. This work should be done in association with the local inhabitants and the Department of Horticulture and Watershed Management. This will also improve the economic base of the local inhabitants.

5.2.2.4 Development of Nurseries

It includes development of nurseries of broad leaved plants, fruits, herbs and grasses etc for conducting social forestry, community/ individual horticultural and herbal gardening programmes in the recharge zones. This work should be done in association of the local community and the Department of Horticulture and forest.

6.0 MICRO-PLANS FOR SPRINGS/STREAMS REJUVENATION

Micro-plan maps (Fig.12,13,14 and 15) of the experimental area depict the notional locations/sites of mechanical and biological treatment measures. This map may work as a faithful guide for implementation of micro-plans in field. A brief account micro-plan related to proposed mechanical and biological treatments is presented in the following paragraphs.

6.1 Mechanical Treatment

Based on the standards of different mechanical treatment measures, geomorphologic and geographical characteristics numbers of each mechanical treatment measure were worked out which are presented in Table-4. A brief account of each mechanical treatment measure is described in the following paragraphs.

Table-4: Estimated numbers of mechanical treatment measures to be constructed in the experimental recharge area.

Name of Mechanical Measures	Number of Treatment Mechanical Measures			
	Forest Land	Barren Land	Abandoned Agricultural Land	Agricultural Land
Infiltration Holes	90000	160000	150000	
Infiltration Trenches	291	2050		
Bio-plugs				
Rill/Gully –Plugs	5			
Terrace Ponds				
Contour Bunds				
Check Dams				
Rainwater Harvesting for domestic purpose				
Rainwater Harvesting for groundwater augmentation				

6.1.1 Infiltration Holes

Infiltration holes are small holes (size 10-15 cm deep, 5cm dia and density 5 per/m²) as shown in Plate-V. The purpose of these infiltration holes is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These infiltration holes should be constructed in barren land, forest land and also in abandoned agricultural land. Figure10 depicts the notional locations of these proposed infiltration holes which have been to be constructed as much as 90,000 in forest land, 160,000 in barren land, and 1,50,000 in abandoned agricultural land (Table-4). These holes may be filled after 2 monsoon rains, then new holes have to be constructed. This work should be done in association with department of Watershed Management.

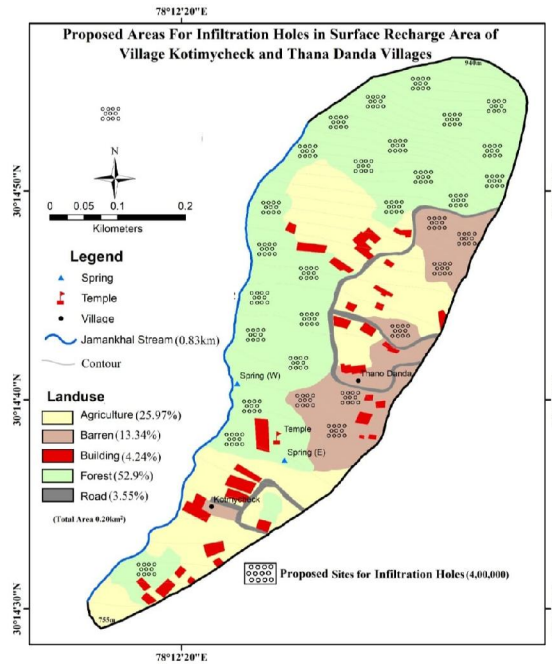


Fig.9:Location of proposed sites for Infiltration Holes.

6.1.2 Infiltration Trenches

Infiltration trenches are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in *Plate-VI*, on the hill slopes made at a spacing of 10m at an average slope condition of 20° . The distance of spacing may be decreased in case the hill slope is low. These should be constructed in stagger pattern. The purpose of these infiltration trenches is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These trenches have to be constructed in the forest and barren lands of the study preferably on convex hill slopes of barren land, degraded forest (Fig.11). A total of about 2341 trenches have to be constructed, i.e., 291 in forest land and 2050 in barren land. These trenches may be filled after 3-4 monsoon rains. After filling completely, a plant should be planted in this filled trench and new trench should be constructed in place of the filled trenches. This work should be done in association with department of Watershed Management.

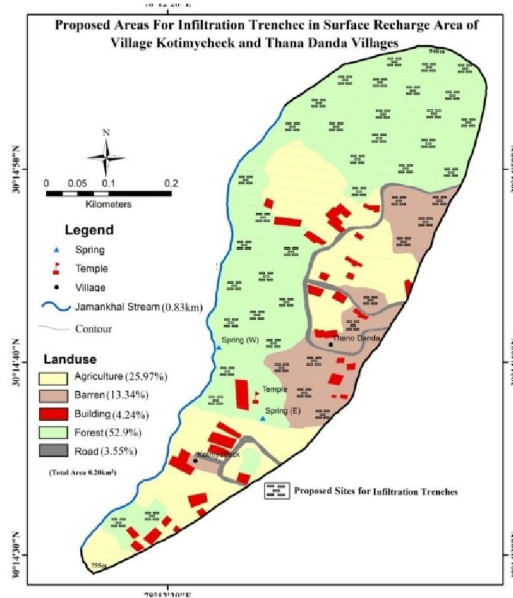


Fig.10:Location of proposed sites for Infiltration Trenches.

6.1.3 Bio- Plugs

These are small bio-plugs to be constructed by vegetation and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rain water to facilitate it for groundwater recharge (Plate-VII). The size of these bio-plugs depends upon the local terrain condition. In the study area, there are two rills/gullies. About 20 Bio-Plugs at an vertical interval shall be needed across in the upper reaches of both the rills/gullies (Fig.12). This work should be done in association with department of Watershed Management and Soil Conservation.

6.1.4 Rill Plugs/Gully Plugs

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep gullies and rills are being developed. To control this process of gullying/rilling, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of streams, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-VIII) depends upon the local terrain condition. In the study area 5 gully plugs shall be needed in the middle reaches across 3 ephemeral channels (Fig.12). This work should be done in association with department of Watershed Management and Soil Conservation.

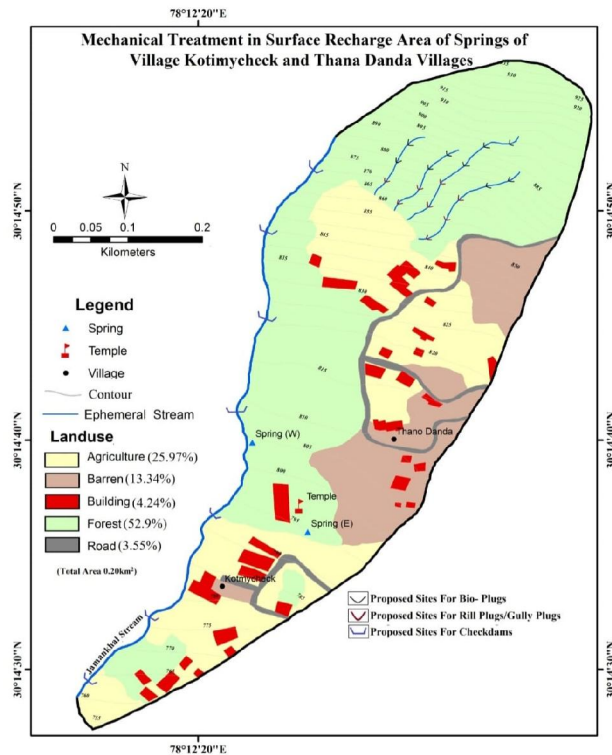


Fig.11:Location of proposed sites for Bio-Plugs, Rill-Plugs and Checkdams.

6.1.5 Contour Bunds

The contour bunds are earthen embankments along contours to capture rainwater in agricultural land (Plate-IX). These help in storing the rainwater for infiltration to groundwater augmentation, reduce erosion and also help in irrigating agricultural crops. These contour bunds should develop at the edge of agricultural land in association with the local residents. As per rough estimate about 5000 running meters contour bunds shall be needed in the study area. Figure 13 depicts the notional locations of these contour bunds.

6.1.6 Terrace Ponds

Terrace ponds are small, artificial water tanks constructed on terraced fields in gentle hill slope areas. These are small shallow tanks with earthen embankments (Plate-X). Their size varies, but typically 10-20 m² in area and 1-2 m deep. The excavated soil of the dig pond are should be used for construction of embankments. They are designed to collect and store rainwater through nearby channels or pipes bring runoff from surrounding catchment areas runoff, providing water for recharge in to the aquifer, irrigation and other uses. They also help to control erosion in the hills. In the barren land 10 and in the agricultural 20 terrace ponds may be constructed in in the study area. Figure 13 depicts the suitable sites for construction of terrace ponds in the study area which may be constructed in association with the local inhabitants. .

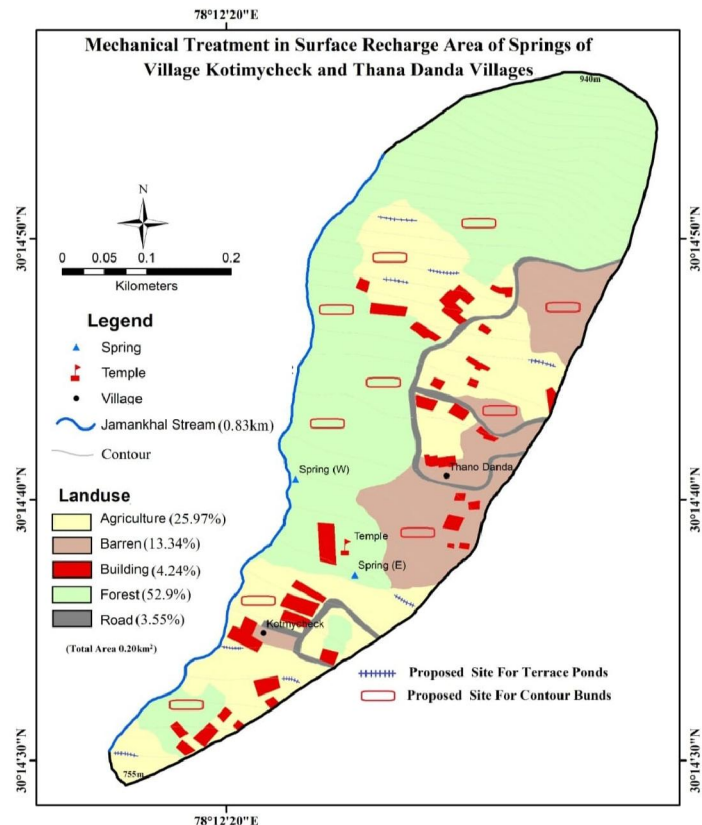


Fig.12:Location of proposed sites for *Contour Bunds and Terrace Ponds*.

6.1.7 Check Dams

These are series of small check dams, constructed using cement and iron across second/third order streams having a height of 1.0 m to 1.5m (depending upon the local terrain condition of the site) (Plate-XI). The purpose these check dams is to check the flow of intermittent/seasonal streams and to store therein water to facilitate it for groundwater recharge, irrigation and to mitigate the process of erosion. In the study area two such check dams may be constructed at the mouth of two rills/gullies (Fig. 12), if needed absolutely

6.1.8 Rooftop rainwater harvesting

Fig. 16:Rooftop rainwater harvesting structures for domestic uses and for groundwater recharge.

6.2 Biological Treatment

6.2.1 Afforestation and Plantation

Afforestation means establishing a forest in an area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, promote biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in the study area:

1. about 0.032 km² land which constitutes about 18.12% of the study area is under barren condition. The spatial distribution of this barren land is presented in figure 6. This entire barren land may be used for i) afforestation of water harvesting (broad leaved) species, local shrubs and grasses such as Napier and vertebrae; and ii) plantation of a single species (monoculture) and non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department.
2. about 0.032km² land of the study area which most of the part is as abandoned barren condition (Fig.7) which may be used for individual and also community based horticultural farming in association with local community. This work should be done in association with the Department of Horticulture.
3. about 0.132km² land is under agricultural and barren condition in which on suitable sites nurseries of: i) broad leave plants, local shrubs and grasses; ii) different herbal and ornamental plants; iii) different fruits; and flowers for floricultural development can be done in association with department of Forest and Horticulture.

should be done by initiating massive social forestry programmes in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.

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